Hyperparameter Optimisation: A Comparative Study of Five Algorithms on YAHPO Gym Benchmarks

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Hyperparameter Optimisation for Machine Learning — Assignment 1

1 Introduction

Hyperparameter optimisation (HPO) is a fundamental problem in machine learning: given a learning algorithm and a task, find the configuration of its hyperparameters that maximises predictive performance. We evaluate five HPO strategies — Random Search (RS), Grid Search (GS), Successive Halving (SH), Hyperband (HB), and Bayesian Optimisation (BO) — on two multi-fidelity benchmarks from YAHPO Gym [5]: NB301, a neural architecture search problem on CIFAR-10, and rbv2_xgboost, an XGBoost hyperparameter tuning problem. Our results show that HB achieves the best final performance on the high-dimensional categorical NB301 space, while multi-fidelity methods (SH, HB) exhibit a pronounced early-incumbent advantage on rbv2_xgboost due to the high correlation between cheap and expensive evaluations.

2 Methods

Random Search (RS) samples configurations uniformly at random and evaluates each at maximum fidelity [1]. It does not learn from observations and serves as the baseline.

Grid Search (GS) exhaustively evaluates a finite discretisation of the search space, using 5 evenly-spaced points per continuous dimension and all choices for categorical dimensions; when the resulting grid exceeds 10,000 configurations (as in the 34-dimensional categorical NB301 space), configurations are drawn uniformly from the per-dimension grids. GS does not adapt to observed results.

Successive Halving (SH) [2] is a multi-fidelity algorithm that begins with n random configurations at minimum budget $r_{\min}$, evaluates all, keeps the top $\lfloor n/\eta \rfloor$, and promotes them to budget $r_{\min} \cdot \eta$, repeating until one configuration reaches maximum budget $r_{\max}$. We use $\eta = 3$ and $n = \lceil \eta^{s_{\max}+1}/(s_{\max}+1) \rceil$, where $s_{\max} = \lfloor \log_\eta(r_{\max}/r_{\min}) \rfloor$.

Hyperband (HB) [3] addresses SH's sensitivity to the initial choice of n by running $s_{\max}+1$ parallel brackets of SH, each with a different (n_s, r_s) trade-off derived from the formula $n_s = \lceil (B/r_{\max}) \cdot \eta^s/(s+1) \rceil$, $r_s = r_{\max} \cdot \eta^{-s}$. The most exploratory bracket (s=$s_{\max}$) evaluates many configurations cheaply; the least exploratory (s=0) is equivalent to RS at full budget.

Bayesian Optimisation (BO) [4] fits a Gaussian Process (GP) surrogate with a Matérn-5/2 + WhiteKernel to all observations and selects the next configuration by maximising Expected Improvement (EI). We implement both EI and its optimisation from scratch using NumPy only: after 5 random warm-start evaluations, BO evaluates EI for 1,000 random candidates per iteration and returns the best. BO always evaluates at maximum fidelity; the Matérn-5/2 kernel is chosen for its balance of smoothness and flexibility [4].

3 Experimental Setup

We benchmark all algorithms on two YAHPO Gym scenarios: (1) NB301 / CIFAR-10: fidelity = training epochs ($r_{\min}$=1, $r_{\max}$=98), metric = validation accuracy; (2) rbv2_xgboost / instance 16: fidelity = training-set fraction ($r_{\min}$ and $r_{\max}$ read from benchmark metadata), metric = accuracy. Each algorithm receives the same total budget of $B = 50 \times r_{\max}$ fidelity units, giving RS and BO approximately 50 full-fidelity evaluations, while SH and HB can explore more configurations cheaply within the same budget. Budget is measured in fidelity units consumed — the standard comparison protocol for multi-fidelity HPO [3]. Hyperparameter spaces are accessed via ConfigSpace without the forbidden methods (sample_configuration, get_array, generate_grid); all sampling is implemented manually. All experiments are repeated for 10 independent random seeds; we report mean ± standard deviation of the best incumbent found.

4 Results

Table 1, Figure 1, and Figure 2 summarise our results.

NB301 (CIFAR-10). All methods reach approximately 93% validation accuracy within the first 500 budget units (Figure 1, left), but thereafter diverge.

Table 1: Final incumbent performance (mean $\pm$ std over 10 seeds). Best per benchmark in bold.

Algorithm	NB301 val_accuracy	rbv2_xgb accuracy
Random Search	93.73 ± 0.17	0.962 ± 0.005
Grid Search	93.71 ± 0.18	0.963 ± 0.010
Succ. Halving	92.79 ± 0.17	$\mathbf{0.968 \pm 0.007}$
Hyperband	$\mathbf{94.16 \pm 0.20}$	0.967 ± 0.007
Bayesian Opt.	94.08 ± 0.32	0.966 ± 0.013

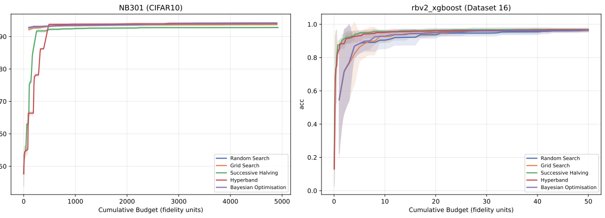

Figure 1: Incumbent curves (mean $\pm$ 1 std over 10 seeds). Left: NB301/CIFAR-10. Right: rbv2_xgboost/instance 16. Budget is measured in fidelity units consumed.

HB achieves the highest final accuracy (94.16 ± 0.20), followed closely by BO (94.08 ± 0.32). RS and GS form a mid-tier baseline ($\approx$93.7%), while SH is the weakest (92.79 ± 0.17). HB's advantage over SH stems from its bracket portfolio: a single SH bracket exhausts its initial candidate set and restarts from scratch, whereas HB concurrently maintains brackets at different fidelity trade-offs, sustaining exploration of promising regions throughout the budget. BO's high variance (± 0.32, Figure 2) reflects the difficulty of fitting a GP to NB301's 34-dimensional categorical space — the one-hot encoding produces a 234-dimensional input, degrading surrogate quality under the curse of dimensionality. RS and GS perform nearly identically, confirming that non-adaptive methods cannot exploit evaluation history regardless of the search strategy.

rbv2_xgboost (Dataset 16). Performance differences are smaller overall (range 0.962–0.968), but the incumbent curves (Figure 1, right) reveal a pronounced early advantage for multi-fidelity methods: SH and HB reach $\approx$ 0.96 accuracy within only 10 budget units, while RS, GS, and BO require 20–30 units to reach comparable performance. This advantage arises because cheap low-fidelity evaluations (small training fractions) are highly informative for XGBoost — the ranking of configurations is stable across fidelity levels, so SH and HB identify and promote the best candidates early. By budget exhaustion all five methods converge to similar final accuracy, consistent with this benchmark's smooth, well-conditioned objective surface.

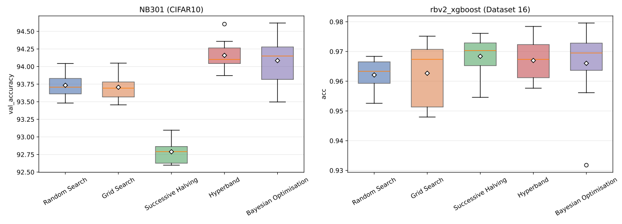

Figure 2: Distribution of final incumbent across seeds. HB achieves the highest median on NB301; SH and HB tie on rbv2. BO exhibits the largest spread on NB301 (outlier visible).

5 Conclusions

HB is the most robust algorithm across both benchmarks, achieving the highest accuracy on NB301 and matching SH on rbv2_xgboost, with consistently low variance. Its bracket portfolio effectively amortises the risk of a suboptimal fidelity trade-off — the key advantage over a single SH bracket. BO is competitive on NB301 but its GP surrogate degrades in high-dimensional categorical spaces, yielding higher variance. Multi-fidelity methods (SH, HB) are particularly effective when cheap proxy evaluations are informative, as demonstrated by their early-incumbent advantage on rbv2_xgboost. RS and GS are reliable but suboptimal baselines; practitioners should prefer HB or BO for any budget exceeding a few dozen full evaluations.

References

[1] J. Bergstra and Y. Bengio. Random search for hyper-parameter optimization. Journal of Machine Learning Research, 13:281–305, 2012.

[2] K. Jamieson and A. Talwalkar. Non-stochastic best arm identification and hyperparameter optimization. In AISTATS, 2016.

[3] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, and A. Talwalkar. Hyperband: A novel bandit-based approach to hyperparameter optimization. Journal of Machine Learning Research, 18(185):1–52, 2018.

[4] J. Snoek, H. Larochelle, and R. P. Adams. Practical Bayesian optimization of machine learning algorithms. In NeurIPS, 2012.

[5] F. Pfisterer, L. Schneider, J. Moosbauer, M. Binder, and B. Bischl. YAHPO Gym – An efficient multi-objective multi-fidelity benchmark for hyperparameter optimization. In AutoML Conference, 2022.